Our Frist Python Program

```
print("Hello World")
```

```
Hello World
```

```
print('Hello World')
```

```
Hello World
```

```
print('Inception BD')
```

```
Inception BD
```

```
print(Inception BD)
```

```
  File "/tmp/ipython-input-2043638736.py", line 1
    print(Inception BD)
          ^
SyntaxError: invalid syntax. Perhaps you forgot a comma?
```

```
print(9)
```

```
9
```

```
print("9")
```

```
9
```

```
print(9.9)
```

```
9.9
```

```
print(True)
```

```
True
```

```
print(False)
```

```
False
```

```
print("Hello",1,2,3,4,5,6.6,True)
```

```
Hello 1 2 3 4 5 6.6 True
```

```
print("Hello",1,2,3,4,5,6.6,True, sep="-")
```

```
Hello-1-2-3-4-5-6.6-True
```

```
print("Hello",1,2,3,4,5,6.6,True, sep=",")
```

```
Hello,1,2,3,4,5,6.6,True
```

```python
print("Bappy", end="-")
print("Inception")
```

```
Bappy-Inception
```

```python
print("Bappy", end=" ")
print("Inception")
```

```
Bappy Inception
```

Data Type in Python & Comments

1. Integer
2. Float (decimal)
3. Boolean
4. String (text)
5. Complex
6. List
7. Tuple
8. Sets
9. Dictionary

```python
# This is a single line comment, this print function prints a sentence
print("My name is Bappy")
```

```
My name is Bappy
```

```python
"""
This is Multiline Comment
this print function prints a sentence
fhsdjhfksdhfkldsf
sdkhfkdjshflkdsf
sdlhfjdlkshfkdjs
"""

print("My name is Bappy")
```

```
My name is Bappy
```

```python
# Integer

print(2)
```

```
2
```

```python
print(200000000000)
```

```
200000000000

type(200000000000)

int

# Float

print(3.3)

3.3

type(3.3)

float

# Bolean
# Python is case sensative language

print(False)

False

type(False)

bool

# String

print("Bappy Inception")

Bappy Inception

type("Bappy Inception")

str

# Complex

print(5 + 7j)

(5+7j)

type(5 + 7j)

complex

# List

print([1,2,3])

[1, 2, 3]

type([1,2,3])
```

```
list

# Tuple

print((1,2,3))

(1, 2, 3)

type((1,2,3))

tuple

# Sets

print({1,2,3})

{1, 2, 3}

type({1,2,3})

set

# Dictionary

print({"Name":"Bappy", "Age":25})

{'Name': 'Bappy', 'Age': 25}

type({"Name":"Bappy", "Age":25})

dict
```

Variables, Keywords & Identifiers in Python

Variables

```
a = 2  # int
print(a)

2

b = 3.3 # Float
print(b)

3.3

c = True
d = 'Bappy'
```

```
print(c)
print(d)
```

```
True Bappy
Bappy
```

```
type(c)
```

```
bool
```

```
name = "Alex"
print("Welcome", name)
```

```
Welcome Alex
```

```
a = 2
b = 5

print(a+b)
```

```
7
```

```
# Dynamic Typing
a = 7

# Static Typing
int a = 7;

type(a)
```

```
int
```

```
a = 7
a = "bappy"
print(a)
```

```
bappy
```

```
# Dynamic Binding
a = 5
print(a)
a = "bappy"
print(a)

# Static Binding

int a = 7;

str a = "bappy";
```

```
5
bappy
```

```
a = 1
b = 2
c = 3

print(a)
print(b)
print(c)
```

```
1
2
3
```

```
a = 1
b = 2
c = 3

print(a, b, c)
```

```
1 2 3
```

```
a,b,c = 1,2,3
print(a, b, c)
```

```
1 2 3
```

```
a = 5
b = 5
c = 5

print(a, b, c)
```

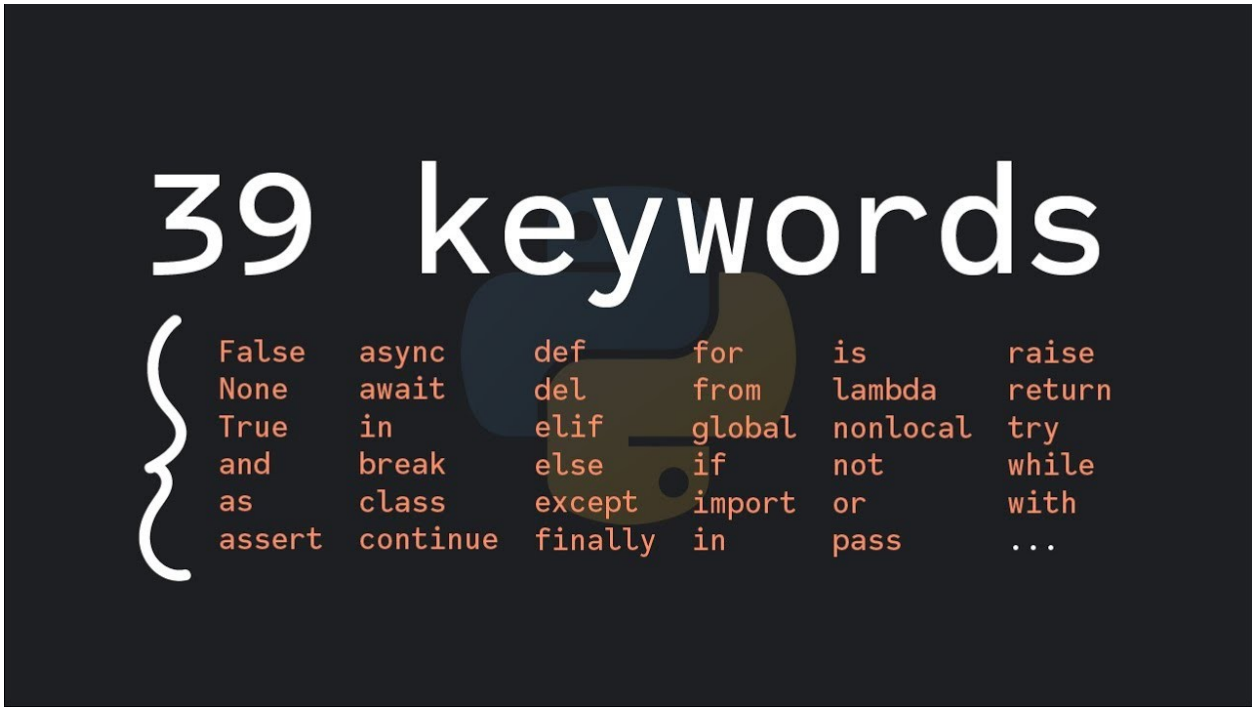
```
5 5 5
```

```
a = b = c = 5

print(a, b, c)
```

```
5 5 5
```

Keywords



Identifiers

```
name = "Bappy"
print(name)
```

```
Bappy
```

```
# You can't start with any digit

1name = 'bappy'
```

```
  File "/tmp/ipython-input-1109743195.py", line 3
    1name = 'bappy'
    ^
SyntaxError: invalid decimal literal
```

```
name1 = 'bappy'

# You can't use any special chars except _

*name = "Bappy"
```

```
  File "/tmp/ipython-input-1089038123.py", line 3
    *name = "Bappy"
    ^
```

```
SyntaxError: starred assignment target must be in a list or tuple


name$ = "Bappy"

  File "/tmp/ipython-input-3492505726.py", line 1
    name$ = "Bappy"
        ^
SyntaxError: invalid syntax


_name = "Bappy"

name_ = "bappy"

_ = "bappy"
```

Python Input

Static - Calender, Clock

Dynamic - Youtube, Facebook

```
input()

7

{"type":"string"}

var = input()

5.6

print(var)

5.6

type(var)

str

var = float(var)
type(var)

float

var = input("Enter Your Name: ")
print(var)

Enter Your Name: bappy
bappy
```

Simple Calculator using input

```
a = int(input("Enter the first number: "))
b = int(input("Enter the second number: "))


result = a - b
print(result)
```

```
Enter the first number: 5
Enter the second number: 4
1
```

Type Conversion in Python:

1. Implicit - Internally by Python
2. Explicit - By the Programmer

```
# Implicit

a = 5 + 5.5

print(a)
print(type(a))
```

```
10.5
<class 'float'>
```

```
# Explicit

b = 4 + int("4")
print(b)
print(type(b))
```

```
8
<class 'int'>
```

```
num = 34

print(float(num))
```

```
34.0
```

```
num = "34"

print(float(num))
```

```
34.0
```

```
num = 3.4

print(int(num))
```

```
3
```

```
num = 4 + 6j

type(num)

float(num)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipython-input-2729759340.py in <cell line: 0>()
      3 type(num)
      4
----> 5 float(num)

TypeError: float() argument must be a string or a real number, not 'complex'
```

Literals in Python

```
a = 3
print(type(a))
```

```
<class 'int'>
```

```
a = 0b1010 #Binary Literals
b = 100 #Decimal Literal
c = 0o310 #Octal Literal
d = 0x12c #Hexadecimal Literal

print(a)
```

```
10
```

```
print(b)
```

```
100
```

```
print(c)
```

```
200
```

```
print(d)
```

```
300
```

```python
# Float Literals

float_1 = 10.5
float_2 = 1.5e2 # 1.5 * 10^2
float_3 = 1.5e-3 # 1.5 * 10^-3

print(float_1)
print(float_2)
print(float_3)
```

```
10.5
150.0
0.0015
```

```python
#Complex Literal
x = 3.14j

info = """
My name is bappy.
I am teaching Python
I am a Data Scientist
shdflsdhlfsd
sdlkhflksdhnf
hnjklsdjfl
"""

info = "
My name is bappy.
I am teaching Python
I am a Data Scientist
"
```

```
  Cell In[15], line 1
    info = "
           ^
SyntaxError: unterminated string literal (detected at line 1)
```

```python
string = 'This is Python'
strings = "This is Python"
char = "C"
multiline_str = """This is a multiline string with more than one line
code."""
unicode = u"\U0001f600\U0001F606\U0001F923"
raw_str = r"raw \n string"

print(string)
print(strings)
print(char)
print(multiline_str)
```

```
print(unicode)
print(raw_str)
```

```
This is Python
This is Python
C
This is a multiline string with more than one line code.
😀😆
raw \n string
```

```
# True = 1
# False = 0

a = True + 4
b = False + 10

print("a:", a)
print("b:", b)
```

```
a: 5
b: 10
```

```
# None

a = None

x = None
y = 2
z = 4

print(x,y,z)
```

```
None 2 4
```

Operators in Python

- Arithmetic Operators
- Relational Operators
- Logical Operators
- Bitwise Operators
- Assignment Operators
- Membership Operators

Arithmetic Operators

```
print(4+3)
```

```
7
print(4-3)
1
print(4*3)
12
print(4/2)
2.0
print(4//2)
2
print(4%2)
0
print(5**2)
25
```

Relational Operators

```
print(4>5)
False
print(4<5)
True
print(4>=4)
True
print(4<=4)
True
print(4==4)
True
print(4!=4)
False
```

Logical Operators

```
print(0 and 1)
```

```
0
```

```
print(0 or 1)
```

```
1
```

```
print(not 1)
```

```
False
```

```
print(not 0)
```

```
True
```

Bitwise Operators

```
# bitwise and
print(2 & 3)

# bitwise or
print(2 | 3)

# bitwise xor
print(2 ^ 3)

#bitwise not
print(~3)

#right shift
print(4 >> 2)

#left shift
print(5 << 2)
```

```
2
3
1
-4
1
20
```

Assignment Operators

```
a = 2
```

```
a = 2
a += 1
print(a)
3
```

Membership Operators

```
# in/not in
print('B' in "Bangladesh")
True
```